

Chapter 10

Introduction to Free Radicals

Additional Recommended Problems: Chp 10: 19, 20,21,23,25,29,30,32,33

CH 10 Learning Objectives

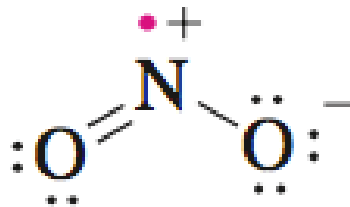
- Describe the structure of an alkyl radical.
- Compare differences in stabilities of alkyl radicals due to substitution.
- Draw the mechanism and predict the products for the halogenation of alkanes.
- Describe why bromination is more selective than chlorination of alkanes in the radical reaction.
- Draw the mechanism and predict the products for hydrohalogenation of alkenes in the presence of peroxides. (anti-Markovnikov addition)
- Determine the products, including stereochemistry, for the metal-ammonia reduction of alkynes.

Free Radicals

- Free radicals are species that contain unpaired electrons
- Some radicals are stable, but most organic radicals are not



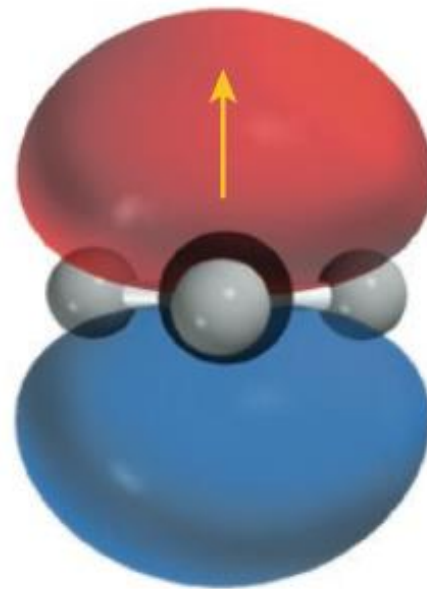
Nitrogen monoxide



Nitrogen dioxide

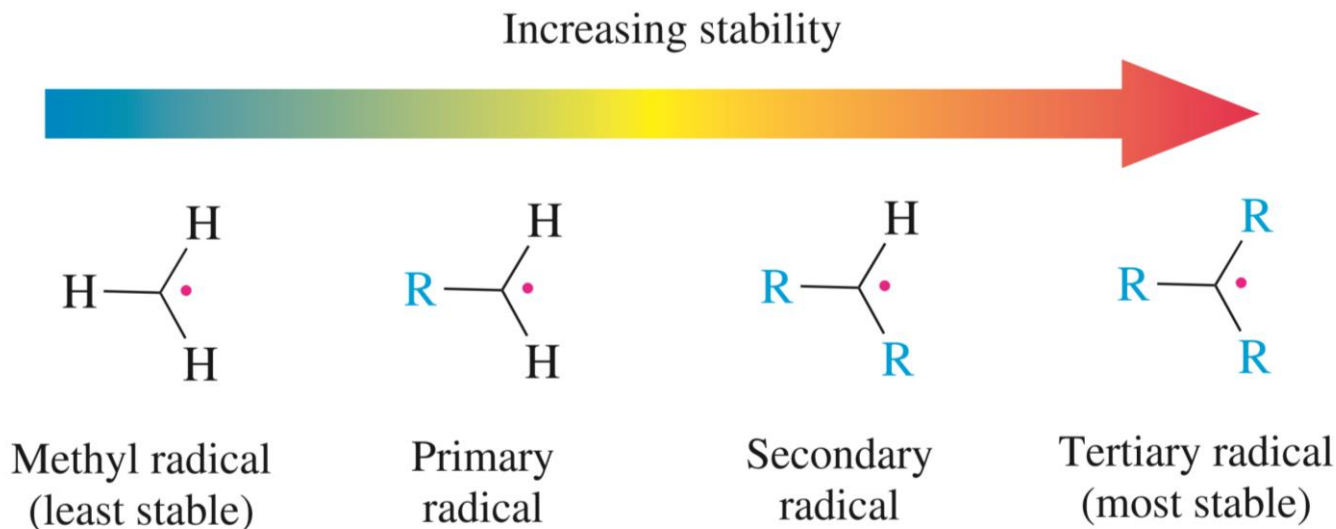
Alkyl Radical Structure

- Alkyl radical contains a carbon with three single bonds and an unpaired electron
- The carbon is sp^2 hybridized

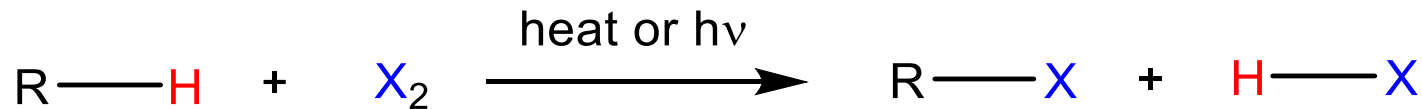


Alkyl Radical Stability

- The more substituted alkyl radical are more stable

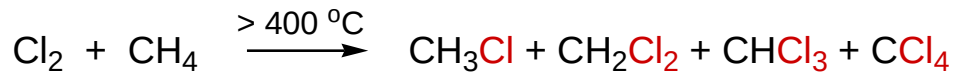


Halogenation of Alkanes



- R-Cl or R-Br from ***alkane***
- F₂ too reactive, I₂ endothermic
- Large E_a barrier to initiate (heat or hν required)

Mechanism of Methane Chlorination



- Radical chain mechanisms (“Fishhook mechanism”) has 3 stages
 1. **Initiation** – generates the free radicals spontaneously
 2. **Propagation** – a radical reacts with an even-electron species to generate even-electron products and new radicals
 3. **Terminations** – radicals combine to quench the reaction

Mechanism of Methane Chlorination (1)

- Overall Reaction: $\text{H}_3\text{C}-\text{H} + \text{Cl}_2 \longrightarrow \text{H}_3\text{C}-\text{Cl} + \text{H}-\text{Cl}$
- **Initiation**
 - Step 1: Dissociation of Cl_2

Mechanism of Methane Chlorination (1)

- Overall Reaction: $\text{H}_3\text{C}-\text{H} + \text{Cl}_2 \longrightarrow \text{H}_3\text{C}-\text{Cl} + \text{H}-\text{Cl}$
- **Propagation**
 - Step 2: Hydrogen atom abstraction from methane by chlorine atom.
 - Step 3: Reaction of methyl radical with molecular chlorine

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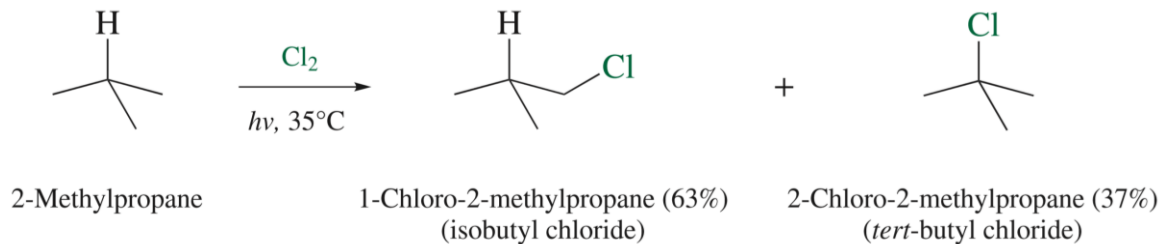
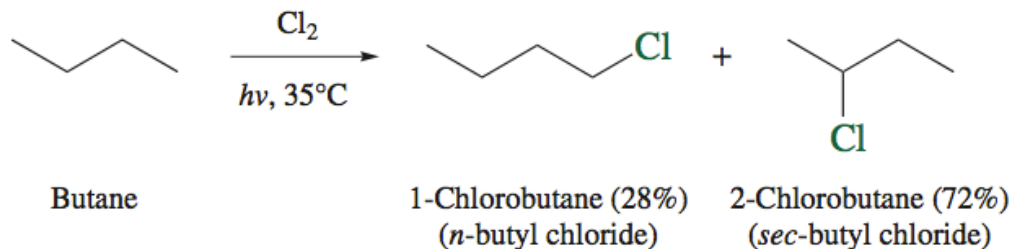
Steps 2 and 3 can repeat many times

Mechanism of Methane Chlorination (2)

- Overall Reaction: $\text{H}_3\text{C}-\text{H} + \text{Cl}_2 \longrightarrow \text{H}_3\text{C}-\text{Cl} + \text{H}-\text{Cl}$
- Termination
 - Two odd-electron species combine

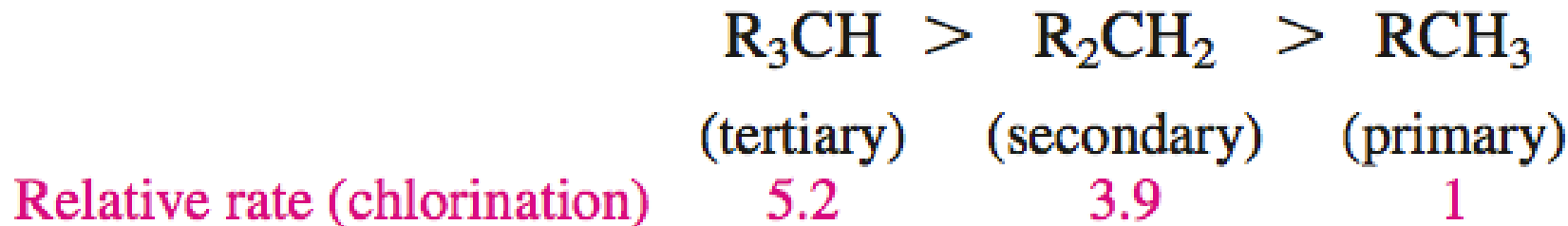
Selectivity of Higher Alkanes

- Chlorine is reactive and gives mixtures of products



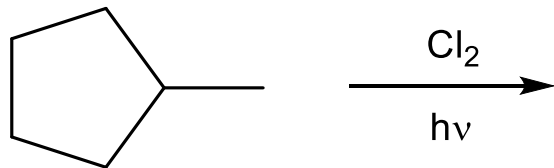
Selectivity of Higher Alkanes (1)

- Chlorine is reactive and gives mixtures of products
- Overall, not very selective



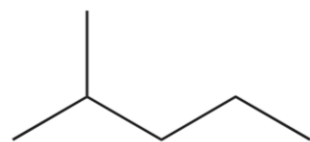
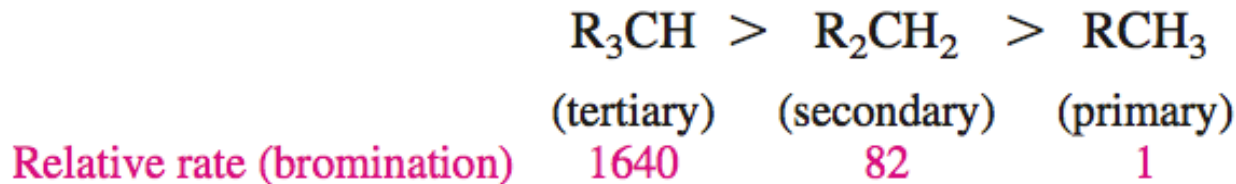
Chlorination Example

- Provide all the constitutional isomers that are produced in the photochemical reaction of methylcyclopentane with chlorine.

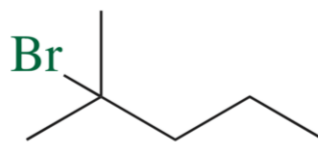
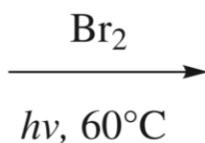


Selectivity of Higher Alkanes (2)

- Bromine on the other hand, is much more selective.

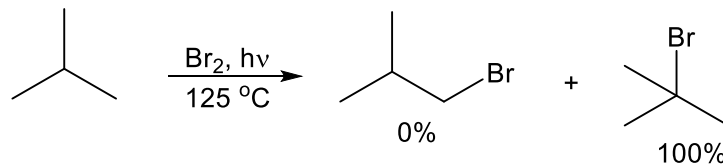
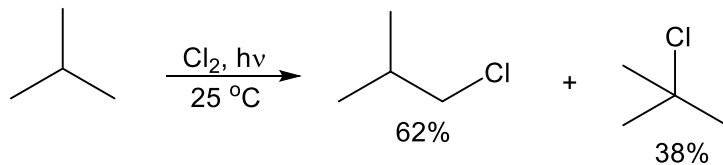
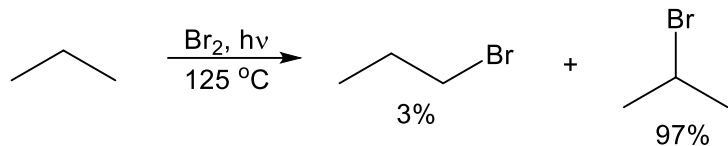
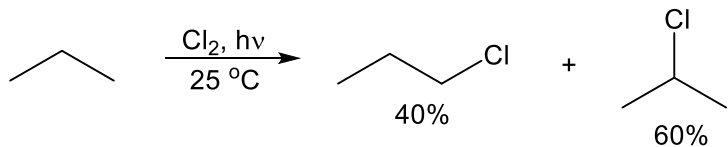


2-Methylpentane

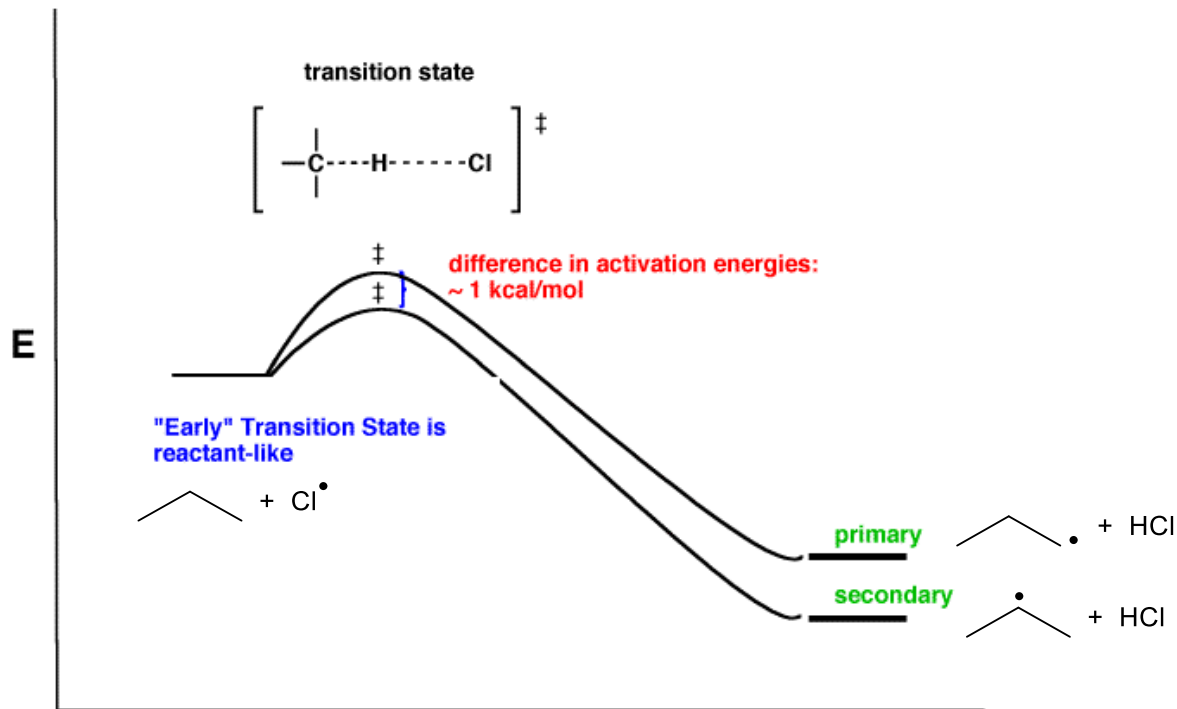


2-Bromo-2-methylpentane
(76% isolated yield)

Selectivity in Halogenation of Alkanes

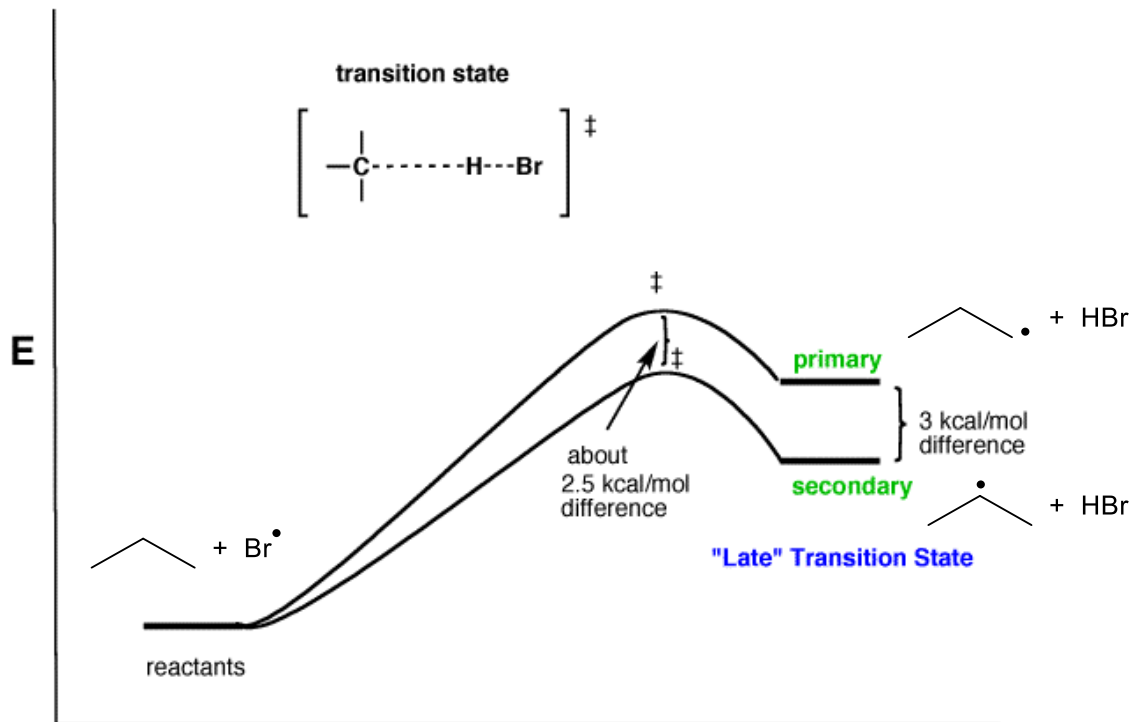


Selectivity in Halogenation of Alkanes



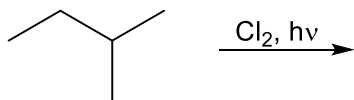
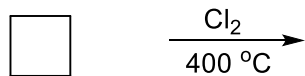
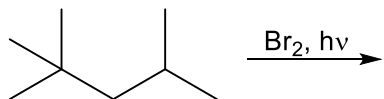
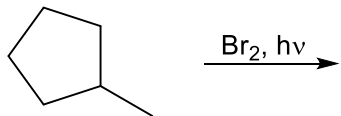
Cl₂ not selective

Selectivity in Halogenation of Alkanes

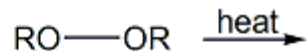
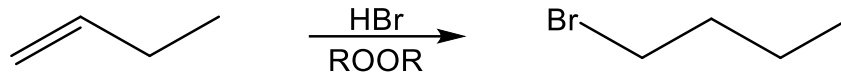


Br₂ selective for most subst. C atom

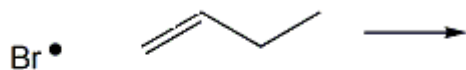
Selectivity in Halogenation of Alkanes



Anti-Markovnikov Addition of HBr

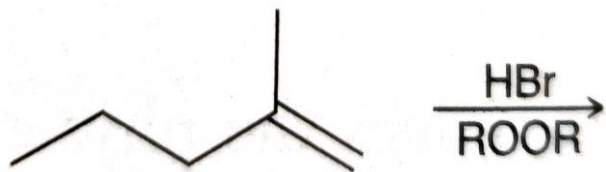
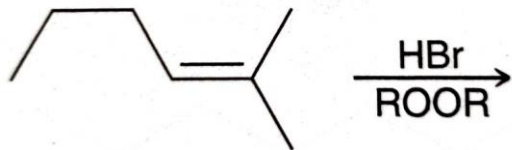


Chain Propagation

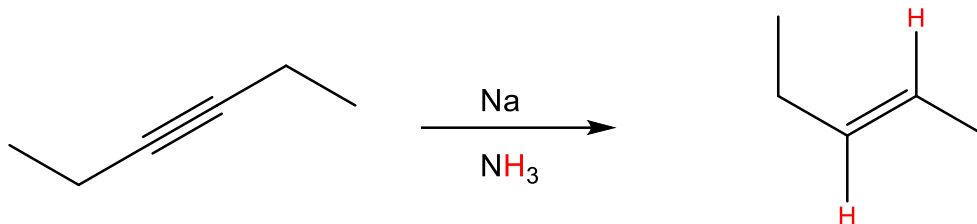


Example:

- Predict the products for the following reactions.

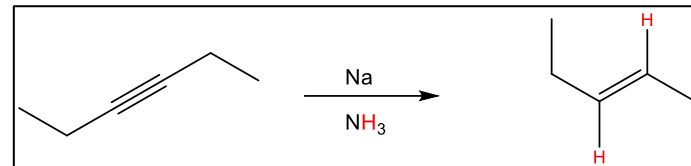


Metal-Ammonia Reduction of Alkynes

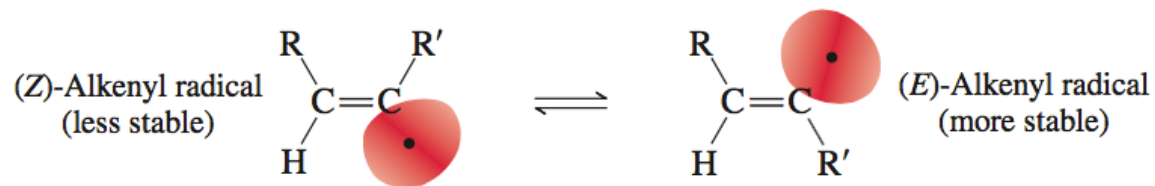
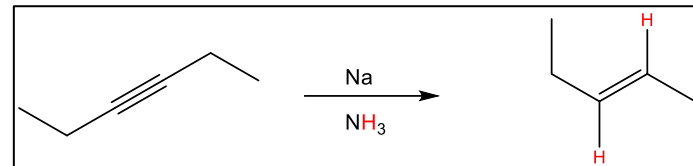


- Sodium metal and liquid ammonia can be used to partially hydrogenate alkynes
- Forms *trans*-alkenes
- Remember, Lindlar's catalyst can be used to reduce alkynes to *cis*-alkenes

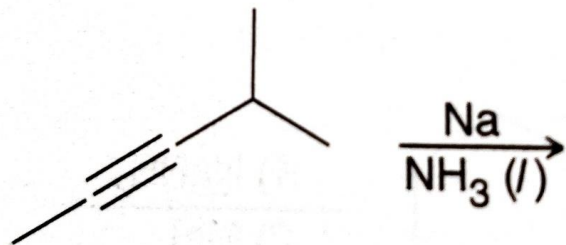
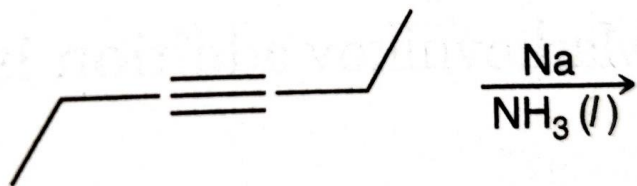
Metal-Ammonia Reduction of Alkynes Mechanism



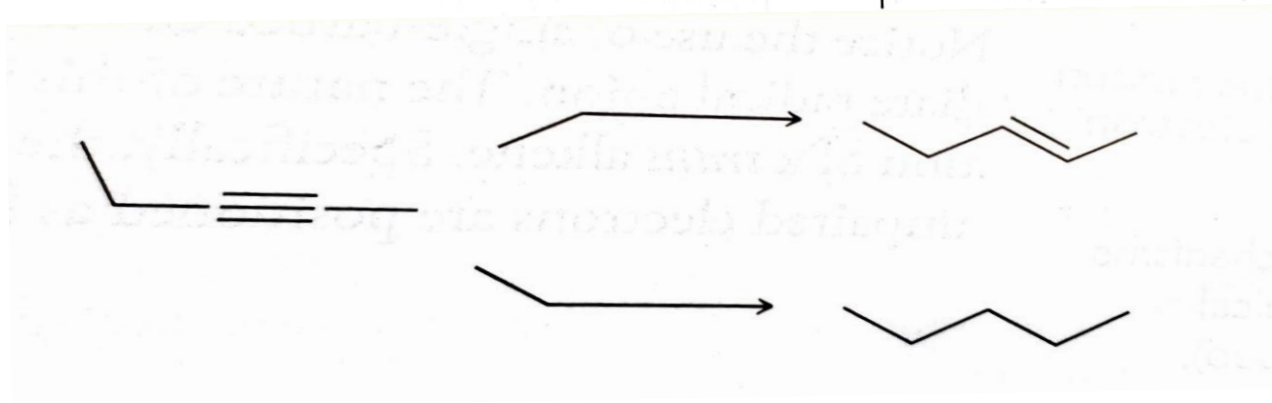
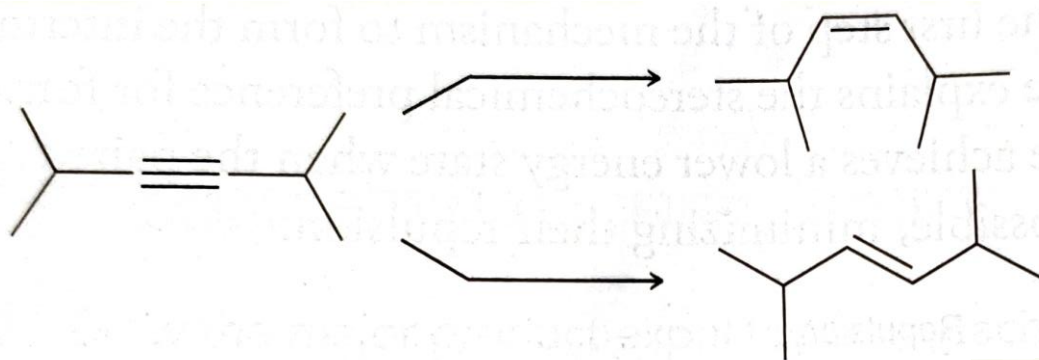
Metal-Ammonia Reduction of Alkynes Mechanism (1)



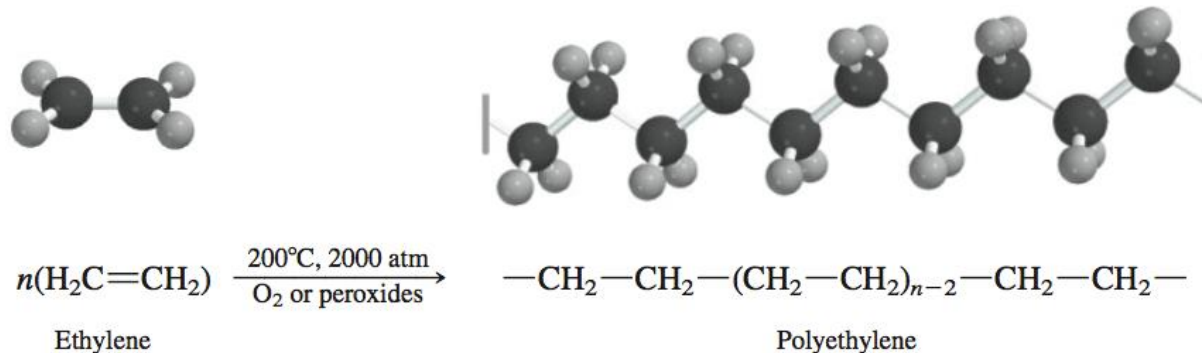
Predict the products



Identify reagents to achieve the following transformations

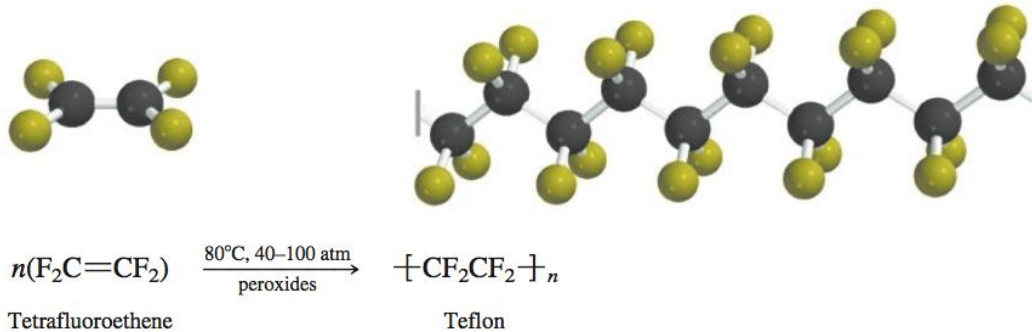


Polymerization via Radical Addition



- Polyethylene (PE) can be generated by the radical polymerization of C_2H_4 .
- Polyethylene is a polymer and the most widely used plastic in the world.

Polymerization via Radical Addition (1)



- Teflon (non-stick polymer) can also be generated by a radical addition